

BITOPI GROUP · IT & ERP

Bitopi Knowledge Assistant

Product Requirements Document — a RAG chatbot over production databases and the PDF library, with a reference for every answer, using a commercial LLM API

Version 1.1 draft for review. Data and indexes at home, the model bought as a service; every claim cited; every user within scope.

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Reviewers: Head of IT, Head of Merchandising, Head of Planning, Data owners (PPM, BIS, HR, MySQL system)

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1. Summary

What we are building

Bitopi Knowledge Assistant is an internal chatbot that answers staff and buyer questions from two sources, live production data (read-only) and a library of 500,000 to 1,000,000 PDFs, and shows for every statement exactly where it came from: the database, view and row for data; the document, page and point for PDFs. Documents, indexes, databases and audit logs stay inside the Bitopi network; the answer itself is written by a commercially licensed LLM reached through an API, which receives only the few retrieved snippets and query rows needed for each question, under a contract that forbids training on our data. Users see only the documents and rows their role, factory or buyer code permits.

The product is defined by three properties that are non-negotiable and shape every other decision: **every claim is cited and every citation can be opened; only the minimum needed for each answer leaves the network, under contract, and every transfer is logged; no user can see data or documents outside their scope.** A fast or clever answer that fails any of these is a defect.

Decisions this PRD is built on

Decision	Choice	Consequence
Data sources	SQL Server ERP databases (Production.PPM, BitopiSplint, HR) and a separate production MySQL system; both read-only	Two SQL dialects; one semantic layer per database; two read-only accounts
Document corpus	0.5–1M PDFs: a mix of per-order documents (tech packs, buyer comments, inspection reports, POs) and company documents (SOPs, policies, manuals)	Order documents must be linked to FileRefID / EO / buyer at ingestion; company documents to factory and department
Model and data policy	Use an existing commercial LLM through an API (no model training or self-hosting); embeddings, reranking, indexes and databases stay on-prem	No GPU servers; a provider contract with no-training and retention terms is on the critical path; per-question snippets and rows are processed by the provider; cost is per token
Users	Internal staff via Active Directory; external buyers via a separate portal seeing only their own orders	Access scope is enforced in retrieval and in SQL, never by the prompt alone
Build path	Learning build in Python (done in the training series); production service in ASP.NET Core with Python workers for ingestion	Prompts, catalogs and tests are shared files used by both

What success looks like

At the end of the pilot, ten staff users ask the assistant real questions for four weeks and it answers at least 85% correctly with valid citations, says 'not found' rather than inventing on the rest, and never shows a user anything outside their scope. At full scale, the same holds for the whole document library and both databases, with a buyer portal in use by at least three buyers, and the operations team can add documents, views and users without a developer.

Key points · The one-paragraph version for management

A chatbot that answers questions from our production databases and from up to a million PDFs, always showing which table or which document page the answer came from, with each user seeing only what they are allowed to see. Our documents and databases stay on our servers; a licensed AI model, bought as a service, writes the answers from the few snippets we send it per question, under a no-training contract. Built in phases over about eight months: a ten-user pilot on a few thousand documents first, then the full library, then buyers.

2. Problem, goals and non-goals

The problem today

Answering an ordinary operational question, 'what did the buyer ask for on this style, and can we still cut on the planned date?', means opening PPM for the date, searching a shared folder for the right comment sheet among several revisions, and reading the SOP to check who must approve a change. Merchandisers and planners do this dozens of times a day. Buyers email to ask for status that is already in our systems. New staff take months to learn where things are. The information exists; finding it and trusting it is the cost.

Goals

1. **Answer with evidence.** Every factual statement in an answer carries a reference the user can open in one click: a document page with the passage highlighted, or the view, row and SQL behind a figure.
2. **Cover both kinds of knowledge.** Live data from the ERP databases and the content of the PDF library, in one conversation, including questions that need both.
3. **Respect scope absolutely.** Staff see their factory's and role's documents and data; buyers see only their own orders. Enforced in the retrieval and data layers, tested adversarially, audited.
4. **Keep data at home, buy the model.** Documents, indexes, databases and audit stay on Bitopi infrastructure. The language model is a commercial API under enterprise terms (no training on our data, limited retention, chosen region); it receives only the snippets and rows needed for each answer, and every transfer is logged.
5. **Be measurably right.** A maintained golden set and automated scoring gate every release; quality is a number, not an impression.
6. **Be operable by IT.** Adding documents, views, users and fixed answers is configuration, not development.

Non-goals

- Writing to any production system. The assistant never inserts, updates, approves or sends; it explains where that is done.
- Replacing PPM, BIS or HR screens. It answers questions; it is not a new front end for transactions.
- General-purpose chat. Questions outside Bitopi's documents and data are declined politely.
- Perfect recall of every PDF from day one. Ingestion is phased; coverage is reported, not assumed.
- Training, fine-tuning or self-hosting a language model. The company has decided to use an existing commercial model through an API. Embedding and reranking models (small, free, run on CPU) are the only models we operate.

3. Users and their questions

Personas

Persona	Who	Daily need	Sensitivity
Merchandiser	Handles buyers, styles and orders for one factory	Buyer comments, approvals, order status, SOP rules for changes	Sees own factory; all buyers of that factory
Production planner	Owens cut plans, PCD, capacity	PCD and ship dates, cut plan progress, escalation rules, planning SOPs	Sees own factory's planning data
Manager / head of department	Oversees several teams or factories	Cross-order summaries, exceptions, policy questions	Sees all factories in their remit
Quality / compliance	Inspections, audits	Inspection reports, SOP compliance, audit trails	Sees QA documents group-wide
HR / admin	Policies, staff matters	HR policies, procedures	Sees HR documents; no order data
External buyer	Buyer's merchandiser or QA	Status of own orders, own comment sheets, inspection results	Sees only documents and rows tagged with their buyer code
IT administrator	Runs the system	Ingestion health, failed documents, question log, evaluation results, user scope	Sees system metadata; can impersonate for support with audit

Representative questions

Question	Source(s)	What the reference must show
What is the PCD for EO 22-0918?	Data (SQL Server)	Database, view, row key, SQL, as-of time
Which TAL orders have PCD next week and are still pending PPM approval?	Data	View, row count, SQL, as-of time; table of rows
What did the buyer ask to change on style TS-4471 in the last comment sheet?	Documents (per-order)	File, revision, page, section, point, quote; link to the EO
Who approves a PCD change after fabric in-house?	Documents (SOP)	SOP code, version, page, section number, quote
Show buyer comments for EO 22-0918 and its current PCD.	Both	One data reference and one or more document references
How many inspection reports for buyer MARCO failed in August?	Data + documents	Query over the document registry (count) with the list of reports as references
What is the stock of fabric lot 88 in the MySQL system?	Data (MySQL)	Database, table/view, SQL, as-of time
Update the PCD to 15 October.	Refused	Explains which PPM screen performs the change and who has authority

Question	Source(s)	What the reference must show
(Buyer) What is the status of my order PO 8812?	Data + documents, buyer-scoped	Only rows and documents tagged with that buyer's code

Access scope by persona

Scope is a set of attributes attached to the user at login and applied as filters everywhere: `factories` (one or more of TAL, RHL, BGL, or ALL), `departments`, `buyer_codes` (for buyers exactly one; for merchandisers all buyers of their factory), `confidentiality_max` (public, internal, restricted). Documents carry the matching attributes in the registry; data views expose the matching columns so row scope can be enforced. Section 7 specifies the mechanism.

4. Functional requirements

Requirements are numbered for traceability. 'Must' items are in the pilot release unless the phase column says otherwise.

FR-1 Ask and answer with references

Q: Which buyer comments apply to EO 22-0918, and what is its current PCD?

The current PCD for EO 22-0918 is 12 Oct 2026 [D1]. The buyer's latest comment sheet asks for the pocket placement to move 1 cm lower on all sizes [P1] and for lab dips to be re-approved before bulk cutting [P2].

[D1] Data reference

Database: Production.PPM (SQL Server)
 Table/view: rag.vw_ExportOrderPCD
 Row key: FileRefID = 4471 (EO 22-0918)
 Query: SELECT PlannedCutDate ...
 WHERE EONo = @eo [show SQL]
 As of: 13 Sep 2026 09:41 · 1 row

[P1] Document reference

File: Buyer comments – EO 22-0918 – Rev2.pdf
 Doc id: 88213 · page 3 · "Fit comments" · point 4
 Quote: "Pocket placement to be lowered by 1 cm on all sizes."
 Linked to: EO 22-0918 · buyer MARCO · TS-4471
 Version: Rev2, 28 Aug 2026 (supersedes Rev1)

Figure 4.1 · Every answer carries references. A data reference names the database, view, row and query; a document reference names the file, revision, page, section and point, with the quoted passage.

ID	Requirement	Priority
FR-1.1	Every factual sentence in an answer carries at least one reference marker ([D1], [P2]). Sentences without a resolvable reference are removed before the answer is shown, or the answer is regenerated once.	Must
FR-1.2	A document reference resolves to: document id, title, revision/version, page number, section heading, point or paragraph number where the document has them, the quoted passage (≤ 300 characters), and a link that opens the PDF at that page with the passage highlighted.	Must
FR-1.3	A data reference resolves to: engine (SQL Server / MySQL), database, schema.view or table, the row key(s) used, the SQL executed (viewable on request), the execution timestamp, and the row count.	Must
FR-1.4	Numbers, dates and codes in an answer must come from a data reference or from a quoted document passage. The system checks this (numeric grounding) and rejects answers that fail.	Must
FR-1.5	When the sources do not contain the answer, the reply is the exact phrase 'I could not find this in the sources available to you', followed by one sentence on where it might be. No partial guess.	Must
FR-1.6	Answers that combine data and documents present them together with references of both kinds.	Must
FR-1.7	Where several document revisions exist, the newest non-superseded one is used by default and the answer says so; older revisions are available on explicit request ('what did Rev1 say?').	Must
FR-1.8	Answers stream to the screen; the references panel appears with the answer.	Should

ID	Requirement	Priority
FR-1.9	Users can ask for the reference list only ('just show me the documents about...') and receive ranked documents with page links and no generated prose.	Should

FR-2 Document ingestion at scale

ID	Requirement	Priority
FR-2.1	Ingest PDFs from designated network folders and, where available, from the ERP document register; detect new, changed and removed files by content hash; re-index only what changed.	Must
FR-2.2	Handle text PDFs, scanned PDFs (OCR, English and Bangla) and mixed layouts; extract tables as tables; strip repeated headers and footers; record OCR confidence per page.	Must
FR-2.3	Link every per-order document to FileRefID, EO number, style, buyer code and factory using, in order: the ERP document register, folder and filename conventions, and fields read from the first page. Record the linking method and confidence; route low-confidence links to a review queue in the admin console.	Must
FR-2.4	Detect document revisions (same order and document type, later date or higher revision) and mark earlier ones superseded.	Must
FR-2.5	Detect exact and near-duplicate files and index once, keeping all locations.	Should
FR-2.6	Chunk by heading and paragraph with overlap; keep page, section heading and point numbers with each chunk; tables as their own chunks with captions.	Must
FR-2.7	Throughput for the initial backfill: at least 50,000 PDFs per day with the planned hardware; steady state: a new or changed document is searchable within one hour.	Must
FR-2.8	Every document, page and chunk is recorded in a registry (SQL Server) with its access attributes, so that 'what is indexed' and 'who may see it' are queryable with T-SQL.	Must
FR-2.9	Failed documents (corrupt, password-protected, unreadable OCR, zero chunks) are listed with reasons in the admin console and can be retried or excluded.	Must
FR-2.10	Removal of a file from the source removes its chunks from search within one hour; legal holds can pin a document.	Should

FR-3 Structured data access

ID	Requirement	Priority
FR-3.1	The assistant reads production data only through dedicated read-only accounts (one per engine) granted SELECT on a dedicated schema of views (rag on SQL Server; a rag_ prefixed schema or view set on MySQL). No access to base tables, no EXECUTE, no linked servers, no cross-database access beyond the views.	Must
FR-3.2	A semantic layer describes each view and column in business terms (what PCD means, that FileRefID joins everything, allowed values), kept as versioned YAML per database and used to build the model's prompt.	Must

ID	Requirement	Priority
FR-3.3	The thirty most common data questions are answered by fixed tools: hand-written parameterised SQL with typed inputs, no model-generated SQL. The list is maintained by IT from the question log.	Must
FR-3.4	Other data questions use model-generated SQL in the correct dialect (T-SQL or MySQL), validated before execution: single SELECT only; deny list (DML, DDL, EXEC, procedures, comments, UNION to system objects, INTO, OPENROWSET, LOAD, INTO OUTFILE); row cap (TOP / LIMIT 500); 10-second timeout; mandatory scope predicates (FR-4.4).	Must
FR-3.5	On SQL error, the error is returned to the model for one corrected attempt; on second failure the user is told the question could not be answered from data and the attempt is logged for review.	Must
FR-3.6	Result sets are rendered as tables in the answer when more than three rows; the SQL is viewable on request; the as-of time is always shown.	Must
FR-3.7	Calculations (days between dates, sums, counts, percentages) are performed in SQL or in code, never by the language model.	Must
FR-3.8	Every executed statement is logged with user, timestamp, tool (fixed or generated), SQL text, row count and duration.	Must

FR-4 Access control

ID	Requirement	Priority
FR-4.1	Staff authenticate with Active Directory; scope (factories, departments, confidentiality level) is derived from AD groups and cached per session.	Must
FR-4.2	Buyers authenticate to a separate portal with individual accounts; scope is exactly one buyer code, plus the document types and views the buyer contract allows.	Must (Phase 3)
FR-4.3	Document retrieval applies the user's scope as a mandatory pre-filter in the vector and keyword indexes; chunks outside scope are never candidates, so they cannot influence ranking, reranking or the prompt.	Must
FR-4.4	Data access applies row scope server-side: on SQL Server via views that read the user's scope from SESSION_CONTEXT (or row-level security policies); on MySQL via scope-parameterised views or predicates injected by the data tool and verified by the guard. A generated query without the scope predicate is rejected.	Must
FR-4.5	The model never receives out-of-scope text; scope is enforced before the prompt is built, not by instructions in the prompt.	Must
FR-4.6	Confidentiality levels: public, internal, restricted. Restricted documents (contracts, pricing, HR files) require explicit group membership and are excluded from buyer scope entirely.	Must
FR-4.7	An adversarial test suite (at least 50 cases) attempts cross-factory, cross-buyer and confidentiality breaches through direct questions, follow-ups and prompt injection in documents. Zero leaks is a release condition.	Must
FR-4.8	Administrators can view and export any user's effective scope and the audit trail of what that user saw.	Must

FR-5 Conversation and language

ID	Requirement	Priority
FR-5.1	Follow-up questions are resolved against the last three turns; the standalone rewritten question is logged with the original.	Must
FR-5.2	English and Bangla questions are accepted; answers are in the language of the question; documents in either language are retrievable from either.	Must (Bangla answers: Should in pilot)
FR-5.3	Out-of-scope requests (write operations, unrelated topics) receive a fixed, polite refusal that names where the task is done.	Must
FR-5.4	Conversations are stored per user for 90 days for the user's own history and for audit; no conversation content is used to train any model.	Must

FR-6 Feedback, audit and administration

ID	Requirement	Priority
FR-6.1	Every answer has thumbs up/down and an optional comment; feedback is stored with the full answer context (question, rewritten question, sources, SQL).	Must
FR-6.2	Audit store records every request: user, scope, question, route taken, tools called, SQL executed, sources returned, answer, model versions, tokens, latency, feedback.	Must
FR-6.3	Admin console shows: ingestion status and throughput, failed documents with reasons, linking review queue, question log with filters, feedback review, evaluation results per release, scope lookup per user.	Must
FR-6.4	IT can add a data view to the semantic layer, add a fixed tool, add a document folder, or change a user's scope through configuration and a redeploy of config only, without code changes.	Must
FR-6.5	Weekly report: questions asked, answered with references, not-found rate, thumbs-down rate, top unanswered questions, top documents cited.	Should

FR-7 Quality gates

ID	Requirement	Priority
FR-7.1	A golden set of at least 150 cases (documents, data, both, not-found, follow-up, adversarial) with expected answers and references, maintained in the repository and grown from the question log.	Must
FR-7.2	An automated evaluation computes: retrieval hit rate, faithfulness, answer correctness, citation validity, not-found handling, scope leaks. It runs on every release candidate and on every model or index change.	Must
FR-7.3	Release rule: no score may fall more than 3 points from the previous release without a written, approved exception; citation validity and scope leaks must be 100% and 0 respectively.	Must

ID	Requirement	Priority
FR-7.4	Monthly manual review of 30 judged cases to confirm the automated judge agrees with human judgement.	Should

FR-8 External model provider controls

ID	Requirement	Priority
FR-8.1	The chat model is a commercial API under a business or enterprise agreement that states: no training on our data; prompt and output retention of 30 days or less (zero where offered); a signed data processing agreement; processing in a named region (Mumbai, Singapore or an EU/US region as approved).	Must
FR-8.2	Data minimisation: a request contains only the system prompt, the (rewritten) question, the top retrieved chunks (≤ 8), and the rows returned by a data tool (≤ 50 rows unless the user asked for a list). Full documents, full tables and the vector index are never sent.	Must
FR-8.3	Embeddings and reranking run on Bitopi servers (CPU is sufficient). Document text is never sent to the provider for embedding.	Must
FR-8.4	Every provider call is logged with: user, chunk ids and row counts sent, token counts, provider, model version, latency, cost. The log answers 'what did we send about EO 22-0918 to the provider?' for any order.	Must
FR-8.5	Traffic to the provider goes through a private endpoint or a fixed egress gateway with TLS; API keys live in the secret store; one key per environment; spending caps and alerts on the provider account.	Must
FR-8.6	The provider is behind one interface; a second, evaluated provider is configured as fallback and can be switched to by configuration within one hour (outage, price change, quality regression).	Should
FR-8.7	Restricted-level documents (HR files, contracts, pricing) are excluded from any request to the provider in the first release, even for users entitled to see them; the answer says the document exists and where to find it.	Must
FR-8.8	Buyer agreements and the privacy notice name the provider as a sub-processor before buyer accounts are enabled.	Must (Phase 3)

5. Non-functional requirements

Scale and sizing

The numbers below plan for one million documents with headroom; the Phase 0 data audit replaces the assumptions with measurements.

Quantity	Planning assumption	Result
Documents	1,000,000 PDFs; 35% scanned	350,000 documents need OCR
Pages	Average 6 pages (order documents short, SOPs long)	6,000,000 pages
Chunks	1.4 per page including tables	≈ 8.5M chunks; plan the vector store for 15M
Vector storage	1,024 dimensions × 4 bytes	≈ 35 GB raw; ≈ 60–90 GB with index; quantisation can halve it
Text and metadata	≈ 3 KB per chunk	≈ 25 GB in the registry
Embedding time (backfill)	bge-m3 on CPU ≈ 25–40 chunks/s per process; 8 processes on a 16-core server ≈ 200–300 chunks/s	≈ 8–12 hours per million chunks; 8.5M chunks ≈ 4–5 days of wall time, run in waves alongside OCR (no GPU required)
OCR time (backfill)	≈ 1 page/s per CPU core with Tesseract	2.1M pages ≈ 25 core-days; 16 cores ≈ 1.5 days of wall time
Parsing and chunking (backfill)	≈ 20 pages/s per core	6M pages ≈ 3.5 core-days
Steady state	500–2,000 new or changed documents per day	Minutes of work per hour on the same hardware
Users	≈ 300 staff, 20 buyer accounts; peak 25 concurrent questions	Sizes the provider quota (requests and tokens per minute) and the API budget
Provider tokens per question	≈ 5,000 in (system prompt + 6 chunks + rows) and ≈ 400 out; plus ≈ 800 in / 60 out for rewrite and routing on a small model	Drives the monthly API cost below

Key idea · API cost and quota, the numbers to watch

Monthly cost $\approx$ questions per month $\times$ (input tokens $\times$ input price + output tokens $\times$ output price). At 3,000 questions per working day, about 66,000 per month, with the token counts above and mid-tier list prices at the time of writing, the chat model costs in the order of USD 1,000–1,800 per month; the small model for rewrite and routing adds under USD 100. Prompt caching of the system prompt and semantic layer typically cuts the chat cost by 30–50%. Prices change; the Phase 0 estimate uses the chosen provider's current sheet and is re-checked every quarter.

Quota: the provider account must allow at least 25 concurrent requests and roughly 150,000 input tokens per minute at peak. Business tiers of the main providers meet this; request the limit at contract time rather than discovering it at go-live.

No GPU servers are required. Embedding and reranking run on an existing 16-core application server; ingestion throughput is CPU-bound and planned in waves.

Performance

Metric	Target (pilot)	Target (full scale)
Document question, time to first token	≤ 3 s (p50), ≤ 6 s (p95)	same
Document question, complete answer	≤ 8 s (p50), ≤ 15 s (p95)	same
Data question (fixed tool)	≤ 4 s (p95)	same
Data question (generated SQL)	≤ 10 s (p95), including one retry	same
Concurrent questions without degradation	10	25
New document searchable	≤ 1 hour	≤ 1 hour
Backfill throughput	n/a	$\geq 50,000$ PDFs/day

Security and privacy

- Documents, indexes, databases, registry and audit run on Bitopi servers. The only outbound traffic is to the contracted LLM provider from the orchestrator host, over TLS to a fixed endpoint (private endpoint where the provider offers one); ingestion workers have no internet access except approved package mirrors during builds.
- Provider terms: business or enterprise agreement with no training on our data, retention ≤ 30 days (zero where offered), a data processing agreement, and processing in an approved region. The agreement is reviewed by management before the pilot and referenced in buyer agreements before Phase 3.
- Read-only database accounts with least privilege; credentials in the organisation's secret store, rotated on staff change; never in code or prompts.
- Documents are treated as untrusted input: text inside sources cannot instruct the model; injection patterns are tested (FR-4.7) and logged.
- Audit logs are retained 2 years, protected from modification, and readable only by IT security and the product owner.
- Buyer portal is separately hosted, rate-limited, MFA-capable, and shares no session with the staff application.

- Personal data in HR documents is restricted-level; the assistant answers HR policy questions but never individual staff records in the first release.

Availability and operations

- Business-hours availability target 99.5%; planned maintenance outside factory shifts.
- Vector store and registry backed up nightly; the vector store can be rebuilt from the registry and source files (the registry is the system of record).
- Monitoring: request latency, error rate, provider latency and error rate, API spend against budget, ingestion backlog, not-found rate, thumbs-down rate; alerts to IT.
- Runbooks for: adding a folder, adding a view, rotating credentials, re-indexing after an embedding model change, restoring from backup, rolling back a model version.

6. Solution architecture

Overview

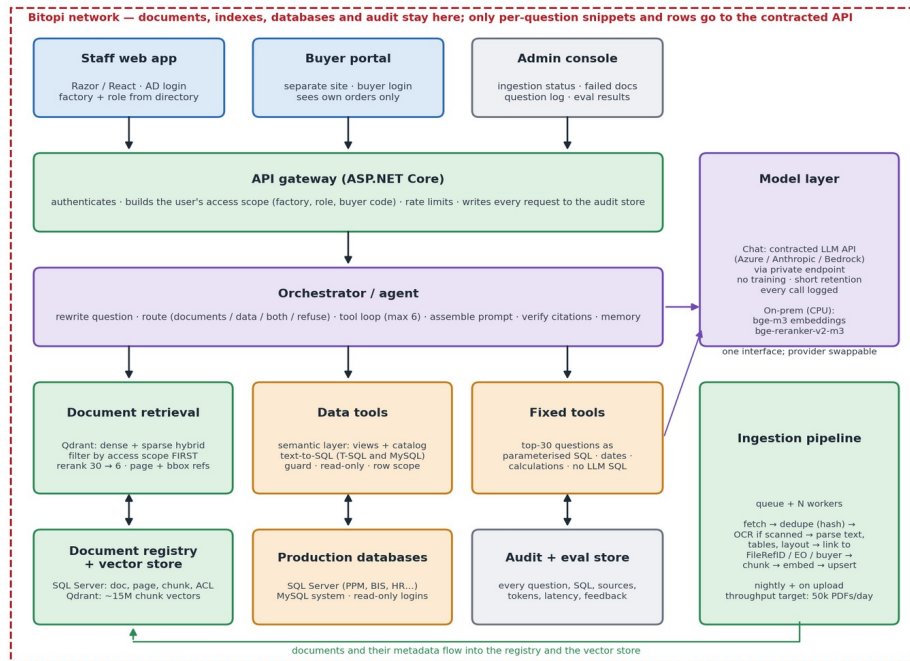


Figure 6.1 · Everything inside the dashed line runs on Bitopi infrastructure; the chat model is the contracted API, reached through the model layer with data-minimisation and logging. Scope is applied at the API gateway and enforced in retrieval and data tools before any prompt is built.

Components

Component	Responsibility	Technology
Staff web app	Chat UI, references panel, open-at-page viewer, feedback	ASP.NET Core Razor Pages or React, hosted on IIS; AD authentication
Buyer portal	Same UI, buyer-scoped, separate host and session	Same stack, separate site and certificate
Admin console	Ingestion, failed docs, linking review, logs, evaluation, scope lookup	ASP.NET Core, IT-only
API gateway	Authentication, scope construction, rate limiting, audit write, request routing	ASP.NET Core minimal API
Orchestrator	Question rewrite, routing, tool loop, prompt assembly, citation verification, memory	ASP.NET Core with Microsoft.Extensions.AI; state machine with a step cap
Document retrieval	Scoped hybrid search (dense + sparse), rerank, return chunks with page/point/bbox	Qdrant (dense + sparse vectors, payload filters); reranker via model gateway
Data tools	Semantic layer, fixed tools, text-to-SQL for two dialects, guard, execution, result shaping	C# with Microsoft.Data.SqlClient and MySqlConnection; TSql ScriptDom and a MySQL parser for validation

Component	Responsibility	Technology
Model provider adapter	One interface for chat completions; routes to the contracted API and to the configured fallback provider; applies data-minimisation rules and logs every call	Microsoft.Extensions.AI IChatClient with provider clients (Azure OpenAI / Azure AI Foundry, Anthropic, or AWS Bedrock); private endpoint
Embedding and reranking service	bge-m3 embeddings and bge-reranker-v2-m3 on CPU for ingestion and for each question	Python service (sentence-transformers or Ollama) on an application server; no GPU
Ingestion pipeline	Fetch, hash, OCR, parse, tables, link, chunk, embed, upsert; retries; metrics	Python workers (PyMuPDF, Tesseract, Docling for hard layouts) on a queue (RabbitMQ or Azure Service Bus on-prem equivalent); scheduled and event-driven
Document registry	System of record for documents, pages, chunks, links, ACL, revisions, ingestion state	SQL Server database KnowledgeRegistry
Audit and eval store	Every request and its context; evaluation runs and results	SQL Server database KnowledgeAudit

The answer path, step by step

1. The gateway authenticates the user and builds the scope object (factories, departments, buyer codes, confidentiality max). It writes the request to the audit store.
2. The orchestrator rewrites a follow-up into a standalone question using the last three turns, then classifies it: documents, data, both, conversation-only, or refuse.
3. For documents: the retrieval service embeds the question locally, runs dense and sparse search with the scope as a mandatory filter, merges with RRF, fetches 30 candidates, reranks to 6 locally, and returns chunks with document id, revision, page, section, point, quote and bounding boxes. Nothing has left the network yet.
4. For data: the orchestrator first tries to match a fixed tool; otherwise it asks the model for SQL in the right dialect using that database's semantic layer, validates the SQL (single SELECT, deny list, row cap, scope predicate present), executes under the read-only account with the user's scope in session context, and shapes the result.
5. The prompt is assembled: system rules, the user's original wording, numbered document sources and data results with their identifiers, and a restatement of the citation rule. The provider adapter checks the data-minimisation limits, logs what is being sent (chunk ids, row counts, tokens), and calls the contracted API. The model writes the answer.
6. Verification: every [D#]/[P#] must map to a supplied source; every number and date in the answer must appear in a data result or a quoted passage. Failure triggers one regeneration with a stricter instruction; a second failure returns 'not found' and logs the case.
7. The answer streams to the UI with the references panel. The audit row is completed with sources, SQL, tokens, latency and model versions. Feedback, when given, is attached to the same row.

Technology choices

Area	Chosen	Considered	Reason
Vector store	Qdrant	SQL Server 2025 VECTOR, pgvector, Milvus	8–15M vectors with mandatory payload filtering and built-in sparse (keyword) vectors; SQL Server 2025's vector indexing is newer at this scale; the registry stays in SQL Server where the team is strongest
Embedding model	bge-m3 (1,024-d, multilingual), run on-prem on CPU	Provider embedding APIs	Keeps the full text of every document inside the network; handles Bangla and long chunks; free per use; fixed for the life of the index
Reranker	bge-reranker-v2-m3, on-prem on CPU	Provider rerank APIs	Matches the embedding family; keeps candidate text inside the network; sub-second for 30 candidates on CPU
Chat model	A current commercial model from the contracted provider, chosen by pilot evaluation on the golden set; a smaller model from the same provider for rewrite and routing	Self-hosted open-weight models on GPU (excluded by company decision)	Strongest available quality for SQL generation and citation discipline; no hardware to buy or operate; swappable behind one interface
Provider	Azure OpenAI / Azure AI Foundry as the default candidate; Anthropic (direct or via AWS Bedrock) as the second candidate evaluated in Phase 0–1	Google Vertex AI	Azure fits the Microsoft estate (Entra ID, private endpoints, existing billing route); Anthropic models are strong on tool use and SQL; both offer enterprise data terms and Mumbai/Singapore regions. Appendix C compares them.
Backend	ASP.NET Core	Python FastAPI	Team skills, IIS/AD integration, existing PPM-Web patterns
Ingestion workers	Python	C#	Best PDF, OCR and ML tooling
OCR	Tesseract (eng+ben) on CPU workers; more workers if throughput demands	Cloud OCR services	On-prem so scanned documents never leave the network; free; Bangla support

7. Data model and access enforcement

Document registry

The registry is the system of record. Everything the vector store holds can be rebuilt from it plus the source files.

Table	Key columns	Purpose
Document	DocumentId, SourcePath, ContentHash, Title, DocType, Revision, EffectiveDate, SupersededByDocumentId, Factory, Department, Confidentiality, BuyerCode, FileRefId, EONo, StyleNo, LinkMethod, LinkConfidence, Status	One row per file; access attributes and links live here
DocumentPage	DocumentId, PageNo, HasTextLayer, OcrConfidence, TextHash	Page-level quality and change detection
Chunk	ChunkId, DocumentId, PageNo, SectionHeading, PointNo, Kind (text/table), Text, BBoxJson, EmbedModel, VectorId	One row per chunk; mirrors the vector store payload
IngestRun	RunId, StartedAt, FinishedAt, Added, Updated, Removed, Failed	Operations history
IngestFailure	DocumentId or SourcePath, Reason, FirstSeen, LastSeen, Resolved	Admin console queue
LinkReview	DocumentId, ProposedFileRefId, Confidence, ReviewedBy, Decision	Low-confidence linking queue

Access attributes and scope

Each document carries Factory (TAL, RHL, BGL, ALL), Department, Confidentiality (public, internal, restricted) and, for order documents, BuyerCode. A user's scope is a set of allowed values for each. A chunk is visible when its document's attributes are all inside the user's scope. This rule is implemented once, as a filter builder, and used by the retrieval service and the keyword index; it is also the rule the adversarial tests attack.

Scope objects (illustrative)

```
-- Staff merchandiser at TAL
factories      = {TAL, ALL}
departments    = {Merchandising, Planning, Common}
confidentiality ≤ internal
buyer_codes    = *    (all buyers of TAL)

-- External buyer MARCO
factories      = {TAL, RHL, BGL}  -- wherever their orders are
departments    = {BuyerFacing}
confidentiality ≤ public
buyer_codes    = {MARCO}         -- exactly one; mandatory on every filter
```

Row scope in the databases

Data access must apply the same scope to rows. On SQL Server, the data tool sets `SESSION_CONTEXT` with the user's factories and buyer codes on the read-only connection, and every rag view reads them: `WHERE Factory IN (SELECT value FROM OPENJSON(SESSION_CONTEXT(N'factories')))` and, for buyer-facing views, `AND BuyerCode = SESSION_CONTEXT(N'buyer_code')`. Row-level security policies are an alternative where views are impractical. On MySQL, which has no session context of this kind, the data tool injects scope predicates into the generated SQL through a parameterised wrapper (`SELECT * FROM (<generated>) q WHERE q.factory IN (?)`), and the guard rejects any statement whose view does not expose the scope columns. Fixed tools take scope as explicit parameters.

Watch out • The one rule that must never be bent

Scope is enforced by filters and predicates the model cannot see or change. If a design proposal ever relies on telling the model 'only show TAL data', it is rejected. The model receives only what the user is allowed to see, and produces text from that.

Reference identifiers

Reference kind	Identifier the UI stores	What the user sees	What 'open' does
Document [P#]	DocumentId, Revision, PageNo, ChunkId, BBoxJson, Quote	File title, revision and date, page, section, point, quote	Opens the PDF viewer at the page with the quote highlighted; shows superseded/current status
Data [D#]	Engine, Database, View, ToolName, SqlText, Parameters, ExecutedAt, RowCount, RowKeys	Database, view, row key(s), as-of time, row count; 'show SQL' link	Shows the SQL and parameters; for single rows, links to the PPM screen for that FileRefID where one exists

8. Delivery plan

Phases and milestones

Phase	Duration	Scope	Exit criteria
0. Discovery and data audit	4 weeks	Inventory the PDF corpus (count, types, scanned share, folders, naming); sample 1,000 files; catalogue candidate views in both databases; define scope attributes and AD group mapping; select the LLM provider (Appendix C), agree contract terms and the payment route, obtain management sign-off on the data-transfer note; set up registry and repository	Audit report with measured numbers replacing Section 5 assumptions; view list agreed with data owners; provider contract signed with no-training and retention terms; data-transfer note approved; golden-set format agreed
1. Pilot (documents + PPM data)	8 weeks	5,000 documents (SOPs, policies, one buyer's order documents); 8 rag views on SQL Server; 10 fixed tools; staff web app with AD; 10 pilot users; golden set of 60 cases; evaluation pipeline	Correctness $\geq 85\%$, faithfulness $\geq 95\%$, citation validity 100%, zero scope leaks in a 30-case adversarial run; pilot users' thumbs-down $\leq 10\%$
2. Scale-out ingestion	8 weeks	OCR farm; linking rules and review queue; revision detection; full backfill to 1M documents in waves (100k, 300k, all); admin console; monitoring	$\geq 95\%$ of documents indexed with a recorded link or classification; backlog < 1 hour; hit rate on a 100-case document set $\geq 90\%$
3. Data expansion and buyers	8 weeks	MySQL semantic layer and views; text-to-SQL for both dialects; row scope in both engines; buyer portal with 3 pilot buyers; adversarial suite to 50+ cases; golden set to 150	Data correctness $\geq 85\%$ on generated SQL, 100% on fixed tools; zero leaks; buyers answer their own status questions without email
4. Hardening and hand-over	6 weeks	Backup and restore drills; fallback-provider switch drill; runbooks; weekly report including API cost; training for IT and champions; Bangla answers; performance tuning to full-scale targets	All non-functional targets met for two consecutive weeks; IT operates for two weeks without developer involvement

Total: about 34 weeks, roughly eight months, with the pilot usable by week 10 because no hardware has to be delivered. Phases 2 and 3 can overlap once the pilot is stable. The dates are estimates for a team of the size below; they tighten or stretch with the provider contract and the state of the document folders.

Team

Role	Load	Responsibility
Product owner (IT / ERP lead)	50%	Priorities, semantic layer with data owners, golden set, acceptance
.NET developer $\times 2$	100%	API, orchestrator, data tools, web apps, admin console

Role	Load	Responsibility
Python / ML engineer	100%	Ingestion pipeline, embedding and reranking service, retrieval tuning, evaluation, provider evaluation
DBA	25%	Read-only accounts, rag views, session context / RLS, registry performance
Infrastructure engineer	20%	Qdrant, queue, embedding service, private endpoint to the provider, monitoring, backups, IIS
Business champions (merchandising, planning, QA)	10% each	Questions, expected answers, pilot feedback

Dependencies

- Provider agreement signed and a working payment route (Microsoft partner billing or an approved USD card) before Phase 1 week 2; the pilot can start on a trial key with masked data while contracting completes.
- Data owners for PPM, BIS, HR and the MySQL system agree the view list and business definitions in Phase 0.
- Document folders are readable by the ingestion service account; an ERP document register, if one exists, is exposed for linking.
- AD groups exist or are created to express factory, department and confidentiality scope.
- Buyer agreements permit portal access to their own documents and status.

9. Success metrics, risks and open questions

Success metrics

Metric	How measured	Pilot target	Full-scale target
Answer correctness	Golden set, LLM judge with monthly human check	≥ 85%	≥ 90%
Faithfulness	Share of claims supported by cited sources	≥ 95%	≥ 97%
Citation validity	Every reference resolves to a real source shown in the prompt	100%	100%
Retrieval hit rate	Expected page in top-6	≥ 88%	≥ 92%
Not-found handling	Not-found cases answered with the exact phrase, no guess	≥ 95%	≥ 98%
Scope leaks	Adversarial suite	0	0
User satisfaction	Thumbs-down rate on answered questions	≤ 10%	≤ 5%
Adoption	Weekly active users / eligible users	10 of 10 pilot users	≥ 60% of eligible staff; ≥ 3 buyers
Coverage	Documents indexed with link or classification / total	n/a	≥ 95%
Freshness	Time from file change to searchable	≤ 1 hour	≤ 1 hour
Latency	p95 complete answer	≤ 15 s	≤ 15 s

Risks and mitigations

Risk	Impact	Mitigation
Order documents cannot be reliably linked to FileRefID / EO	Buyer scope and per-order answers fail; documents become 'general'	Phase 0 audit of naming conventions; ERP register as first source; review queue; unlinked documents excluded from buyer scope and flagged in answers
Poor OCR on old scans	Retrieval misses; wrong codes	Per-page OCR confidence; low-confidence pages flagged; keyword search tolerant matching; prioritise re-scanning of high-value documents
Model invents numbers or misreads tables	Wrong figures with confident tone	Numbers only from data or quoted text (FR-1.4); tables as tables; calculations in SQL; numeric grounding check
Cross-buyer or cross-factory leak	Contractual and trust failure	Scope as pre-filter and row predicate (FR-4); adversarial suite as release gate; audit of everything shown
Text-to-SQL accuracy on a large schema	Wrong data answers	Small, well-described view set; fixed tools for the common 30; guard; retry with error; correctness measured per view; fall back to 'ask IT'

Risk	Impact	Mitigation
Provider dependency: outage, price rise, model retirement or quality regression	Assistant unavailable or worse; cost overrun	Second evaluated provider configured as fallback (FR-8.6); golden set re-run on every model version change; monthly cost report; contract with notice periods on price and model changes
Company data processed by a third party	Contractual, regulatory and buyer-trust exposure	Enterprise terms (no training, short retention, DPA, approved region); data minimisation (FR-8.2); restricted documents never sent (FR-8.7); every transfer logged (FR-8.4); buyer agreements updated before Phase 3
Payment and procurement friction for a foreign API	Pilot delayed by finance rather than technology	Prefer billing through the existing Microsoft partner or a local reseller; start contracting in Phase 0; keep a trial key for the pilot
Duplicate and revised documents	Conflicting answers	Hash dedupe; revision detection; newest-current default; superseded shown on request
Prompt injection through PDFs	Instructions hidden in documents steer answers	Sources treated as data; injection tests in the golden set; citation and grounding checks limit damage
Change management	Staff distrust or ignore the tool	Champions per department; every answer checkable; weekly report shared; fast fixes from thumbs-down
Two database dialects	Double the semantic-layer and guard work	One data-tool interface with dialect plug-ins; start MySQL with fixed tools only, add generated SQL after

Open questions to close in Phase 0

1. Exact size and composition of the PDF corpus: count, distribution by type, scanned share, average pages, folder structure, naming conventions.
2. Does an ERP document register exist that maps files to FileRefID or EO? If not, which conventions can be relied on?
3. Which MySQL system, which tables, and who owns its business definitions?
4. Which AD groups exist for factory and department, and who approves the mapping to scope?
5. Which document types may buyers see, and do buyer agreements need updating?
6. Which confidential document classes must be excluded entirely from indexing in the first release?
7. Which provider and contract tier, and through which billing route; who signs the data processing agreement; which region is approved for processing.
8. Retention rules for conversations and audit, agreed with management and any auditors; whether buyer agreements need an amendment naming the provider as a sub-processor.

Appendix A • Example interactions

Three exchanges showing the reference format in text form as the UI would render it.

User	Who approves a PCD change after fabric in-house, and what is the current PCD for EO 22-0918?
Assistant	A PCD change after fabric in-house requires written approval from the Planning Head on form PCD-02 [P1]. Planners may approve changes of two working days or less if the ship date is unaffected [P2]. The current PCD for EO 22-0918 is 12 Oct 2026 [D1]. References [P1] SOP-PPM-014 PCD Approval, Rev 3 (15 Jan 2026) · page 4 · §3.2 · point 1 · "...requires written approval from the Planning Head..." [P2] SOP-PPM-014 PCD Approval, Rev 3 · page 5 · Table 1 Approval matrix [D1] Production.PPM (SQL Server) · rag.vw_ExportOrderPCD · FileRefID 4471 · fixed tool pcd_by_eo · as of 13 Sep 2026 09:41 · 1 row

User	(external buyer MARCO) What did we ask to change on TS-4471, and has it been done?
Assistant	Your comment sheet Rev 2 asks for the pocket placement to be lowered by 1 cm on all sizes [P1] and for lab dips to be re-approved before bulk cutting [P1]. The order's status shows lab dips approved on 5 Sep 2026 [D1]; I could not find a record confirming the pocket change in the sources available to you. Your merchandising contact can confirm from the internal fit-approval record. References [P1] Buyer comments – EO 22-0918 – Rev2.pdf · page 3 · Fit comments · points 4 and 6 [D1] Order status (buyer view) · EO 22-0918 · as of 13 Sep 2026 10:02

User	Please update the PCD for EO 22-0918 to 15 October.
Assistant	I can't change data. PCD changes are made in PPM › Export Order › Planning tab by a planner and, after fabric in-house, need Planning Head approval on form PCD-02 [P1]. I can show you the current PCD and the approval rule if that helps. [P1] SOP-PPM-014 PCD Approval, Rev 3 · page 4 · §3.2

Appendix B • Choosing the LLM provider

The company has decided to use an existing commercial model rather than train or host one. This appendix gives the comparison and the selection procedure for Phase 0. Product names, regions and terms are as understood in September 2026 and must be confirmed against each provider's current documentation and contract before signing.

Comparison

Criterion	Azure OpenAI / Azure AI Foundry	Anthropic (direct API)	AWS Bedrock (Claude, others)	Google Vertex AI (Gemini)
No training on our data (business terms)	Yes	Yes	Yes	Yes
Retention	Configurable; abuse-monitoring retention can be waived on request	Zero-retention available on enterprise terms	No retention of prompts by default for model invocation	Configurable on enterprise terms
Region choice near Bangladesh	India regions; Singapore	Provider-managed (US); no regional pinning	Mumbai (ap-south-1); Singapore	Mumbai (asia-south1); Singapore
Identity and network fit	Entra ID (our AD), private endpoints, existing Microsoft tenancy	API key; IP allow-listing	IAM; VPC endpoints	IAM; private service connect
Billing route	Existing Microsoft partner / CSP; can avoid foreign card payments	USD card or invoice on enterprise contract	AWS account, USD; local partner possible	Google Cloud account, USD; local partner possible
Model choice	OpenAI models and, via Foundry, others including Claude and Llama	Claude family	Claude, Llama, Mistral, Amazon models	Gemini family plus partner models
Strength for this project	Closest fit to our estate; one vendor relationship	Strong tool use, SQL generation and instruction following; long context	Anthropic models with AWS region and IAM controls	Strong multilingual; competitive pricing
Watch for	Model versions lag the vendor's latest by weeks; quota requests take time	No regional pinning; payment route from Bangladesh	Model availability differs by region	Less used in our team; learning curve

Selection procedure (Phase 0, two weeks)

1. **Contract screen.** Obtain the business/enterprise terms from the two leading candidates (Azure and Anthropic, directly or via Bedrock). Confirm in writing: no training, retention period, DPA, region, notice period for price and model changes, quota available. Eliminate any that fail.
2. **Payment screen.** Confirm with Finance how each would be paid from Bangladesh (partner billing, USD card, invoice). A provider that cannot be paid is not a candidate.
3. **Quality test.** With trial keys and the masked pilot data, run the 60-case golden set through both: document answers with citations, generated T-SQL and MySQL on the semantic layer, Bangla questions, not-found cases, adversarial cases. Record correctness, faithfulness, citation validity, SQL accuracy, latency and cost per question.
4. **Decide.** Choose the primary on quality and fit; configure the other as fallback (FR-8.6). Record the decision, the numbers and the contract references in the PRD's decision log.
5. **Re-evaluate** at every major model version change and at least twice a year; switching is a configuration change plus a golden-set run.

Key idea · What stays the same whatever the provider

Embeddings, reranking, the vector store, the registry, the databases, scope enforcement, the citation model and the audit store are all on-prem and provider-independent. The provider writes answers from snippets we choose to send. Changing provider changes one adapter and one contract; it does not change the system.

Appendix C • Glossary

Term	Meaning
Access scope	The set of factories, departments, buyer codes and confidentiality levels a user may see; applied as filters in retrieval and predicates in SQL.
Chunk	A piece of a document (a section, paragraph or table) that is indexed and retrieved as a unit, with its page and position.
Citation validity	Every reference in an answer points to a source that was actually retrieved and shown to the model.
Faithfulness	The share of an answer's claims that are supported by its cited sources.
Fixed tool	A hand-written, parameterised query for a frequent question; no model-generated SQL.
Golden set	Questions with known correct answers and references used to score every release.
Hybrid search	Combining meaning (vector) search and keyword search; needed for codes such as EO numbers.
Numeric grounding	The check that every number, date or code in an answer appears in a data result or a quoted passage.
OCR	Turning scanned page images into text.
RAG	Retrieval-Augmented Generation: retrieve the relevant sources first, then have the model answer only from them.
Registry	The SQL Server database that records every document, page, chunk, link and access attribute; the system of record.
Reranker	A model that re-scores retrieved candidates against the question to pick the best few.
Semantic layer	Business-language descriptions of the views and columns the model may query, per database.
Superseded	An older revision of a document that is excluded from default answers.
Sub-processor	A third party that processes personal or business data on our behalf; the LLM provider is one and must be named in agreements.
Data minimisation	Sending the provider only the snippets and rows needed for one answer, never whole documents, tables or indexes.